MILITARY SPECIFICATION

FUZE, GUIDED MISSILE, FMU-138/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FMU-138/B Guided Missile Fuze, referred to herein as the fuze.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	Tag, Shipping and Stock.
L-T-100	Tape, Pressure-Sensitive Adhesive, Polyester Film.
TT-I-735	Isopropyl Alcohol.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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MIL-S-6872	Soldering Process, General Specification for.
MIL-T-14757	Trichlorotrifluoroethane.
MIL-C-21723	Composition CH-6.
MIL-C-45662	Calibration System Requirements.
MIL-D-85294	Detonator, Electric, MARK 71 MOD 1.
MIL-S-85354	Switch, Inertia, Fuze.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1246	Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1316	Fuze, Design Safety Criteria for.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MS51938	Seal, Metallic, Lead and Wire.

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PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

Naval Sea Systems Command
(Code Ident 53711)

WS-1639	Detonator, Electric, MARK 85 MOD 0.
OS-8164	Descriptions and Requirements for Lead, Explosive, MARK 9 MOD 0.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

67A87D460	Fusible Link (Dual).
DL 1186AS300	Fuze, Guided Missile, FMU-138/B.
DL 1186AS311	Shipping and Storage Container, Packing and Marking.
SA 497753	Fixture, Explosive Train Drop.
SA 498016	Fixture, Vibration.
SA 498017	Fixture, Jolt.
SA 498018	Fixture, Forty-Foot Drop.
SA 498019	Fixture, Transportation Vibration.
2049577	Cup, Booster.
DL SA2875456	Test Set, Fuze and Air Valve (Tomahawk).
SA 2876062	Fixture, Shock.

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 108	Steel Bars, Carbon, Cold-Finished, Standard Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

NOTE: Drawings and associated documents (such as DLs and LDs) referenced herein which are not listed in Section 2 are delineated on the DLs or LDs cited in Section 2.

3.1 Item description. The fuze was designed in accordance with MIL-STD-1316 and is an electromechanical device that maintains an out-of-line explosive train and an electrical firing circuit in the safe (open) position until certain prerequisite events have occurred after missile launch. The general configuration of the fuze is shown on Figure 1.

3.1.1 Government-furnished property.

3.1.1.1 Autonomous components. When specified in the contract or purchase order, the Government will furnish the contractor with the following components for each fuze produced under a specified contract (see 6.2.1):

a. Four MARK 9 MOD 0 Explosive Leads (OS-8164).

b. One Fuze Inertia Switch (MIL-S-85354).

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c. One Fusible Link, Dual (Drawing 67A87D460).

d. Two MARK 85 MOD 0 Electric Detonators (WS-1639).

e. Two MARK 71 MOD 1 Electric Detonators (MIL-D-85294).

3.1.1.2 Support equipment. When specified in the contract or purchase order, the Government will furnish the independent test facility designated in the contract or purchase order with the following support equipment to perform the environmental tests specified herein (see 6.2.1):

a. One explosive train drop fixture (Drawing SA 497753).

b. One vibration fixture (Drawing SA 498016).

c. One jolt fixture (Drawing SA 498017).

d. One forty-foot drop fixture (Drawing SA 498018).

e. One transportation vibration fixture (Drawing SA 498019).

f. One shock fixture (Drawing SA 2876062).

3.1.2 Government-loaned property. When specified in the contract or purchase order, the Government will loan the contractor and the independent test facility designated in the contract or purchase order one Tomahawk Fuze and Air Valve Test Set (DL SA2875456), referred to herein as the test set (see 6.2.1).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Electrical. The requirements of 3.2.1.1.1 and 3.2.1.1.2 shall be met by the fuze during and after assembly in the configurations shown in Table I. In Tables II, III, and IV, switch position numbers refer to the test set (see 3.1.2), and correspond to the pin numbers in the fuze connector. An electrical schematic for the fuze is shown on Figure 2.

TABLE I. Assembly configurations for electrical requirements.

Configuration	Detonators installed	Welded[1/]	Sealed[1/]
A	No	No	No
B	Yes	Yes	No
C	Yes	Yes	Yes

[1/] See Drawing 1186AS300.

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TABLE II. Resistance (safe condition) requirements for configurations A, B, and C.

Test mode	Ohmmeter scale reading	Switch positions		Resistance				Items checked (see Figure 2)
				Configuration				
				During manufacture		After shipment		
				A, B, and C		C		
HP	Off	HP	C-	Lower limit	Upper limit	Lower limit	Upper limit	
HP1	---	(1-8, 10-12)	9	100 Mohms	∞	100 Mohms	∞	HP
HP2	---	(1-12)	13 1/	100 Mohms	∞	100 Mohms	∞	HP
3469	On	C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	(see Figure 2)
S1	10 Mohms	5	8	2/	2/	2/	2/	C1, R1, 3
S2	100 kohms	All	9	Open	Open	Open	Open	---
S3	100 kohms	7	3	Open	Open	Open	Open	S1E
S4	100 kohms	4	12	Open	Open	Open	Open	Inertia switch
S5	100 kohms	4	1	Open	Open	Open	Open	CR6
S6	100 kohms	1	4	35 kohms	45 kohms	20 kohms	60 kohms	CR6
S7	100 kohms	8	10	27 kohms	38 kohms	25 kohms	41 kohms	R4, 6, 8
S8	100 kohms	11	8	26 kohms	34 kohms	10 kohms	34 kohms	$Q1_E$, R4, 6, 7, 8
S9	1 kohm	6	11	295 ohms	365 ohms	260 ohms	400 ohms	R7, $Q1_C$
S10	100 ohms	11	10	71 ohms	95 ohms	65 ohms	115 ohms	$Q1_C$, R7
S11	1 ohm	4	8	---	---	---	---	---
S11 (Cal) 3/	1 ohm	4	8	---	---	---	---	---
S11 (Net) 4/	---	---	---	0.40 ohm	0.90 ohm	0.30 ohm	1.00 ohm	F2
S12	1 ohm	6	10	---	---	---	---	---
S12 (Cal) 3/	1 ohm	6	10	---	---	---	---	---
S12 (Net) 4/	---	---	---	0 ohm	0.45 ohm	0 ohm	1.00 ohm	S1F

1/ Clip must be attached to case for these tests.

2/ Lower limit will show capacitor (C1) charging. Upper limit (approximately 7 Megohms (Mohms)) may be exceeded, depending on charging time.

3/ Calibration readings are taken at calibration socket.

4/ Net values are the normal readings minus the calibration readings.

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TABLE III. Resistance (armed condition) requirements for configuration A.

Test mode 3469	Ohmmeter scale reading	Switch positions		Resistance		Items checked (see Figure 2)
		C+	C-	Lower limit	Upper limit	
A1	100 kohms	10	5	35 kohms	45 kohms	CR5, S1D
A2	100 kohms	5	10	Open	Open	CR5
A3	1 ohm	7	3	---	---	---
A3 (Cal)[1/]	1 ohm	7	3	---	---	---
A3 (Net)[2/]	---	---	---	0 ohm	0.45 ohm	S1E

[1/] Calibration readings are taken at calibration socket.
[2/] Net values are the normal readings minus the calibration readings.

TABLE IV. Resistance (armed condition) requirements for configurations B and C.

Test mode 3469	Ohmmeter scale reading	Switch positions		Resistance				Items checked (see Figure 2)
				Configuration				
				B		C		
		C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	
A1	100 kohms	10	5	35 kohms	45 kohms	20 kohms	60 kohms	CR5, S1D
A2	100 kohms	5	10	Open	Open	Open	Open	CR5
A3	1 kohm	8	12	500 ohms	700 ohms	500 ohms	700 ohms	Instantaneous detonators S1B, CR3, S1C
A4	1 kohm	12	8	500 ohms	700 ohms	500 ohms	700 ohms	Delay detonators S1A, CR4, F1
A5	1 ohm	7	3	---	---	---	---	---
A5 (Cal)[1/]	1 ohm	7	3	---	---	---	---	---
A5 (Net)[2/]	---	---	---	0 ohm	0.45 ohm	0 ohm	1.00 ohm	S1E

[1/] Calibration readings are taken at calibration socket.
[2/] Net values are the normal readings minus the calibration readings.

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3.2.1.1.1 Resistance (safe condition). The fuze shall meet the resistance requirements of Table II for each of the three configurations (A, B, and C) specified in Table I while in the safe condition (see 6.3.1).

3.2.1.1.2 Resistance (armed condition). The fuze shall meet the resistance requirements of Table III for configuration A, Table I, while in the armed condition (see 6.3.2). The fuze shall meet the resistance requirements of Table IV for configurations B and C, Table I, while in the armed condition.

3.2.1.2 Mechanical operability. The fuze shall meet the mechanical operability requirements of Table V for each of the three configurations (A, B, and C) specified in Table I. Beginning and ending times are indicated by the beginning and ending switching functions. (The requirements of Table V represent fuze calibration times and not fuze arming times.)

TABLE V. Mechanical operability requirements for configurations A, B, and C.

Test	Pressure applied	Solenoid Vdc		Requirements	
		Configuration			
		A and B (Vdc)	C (Vdc)	A and B	C
All go	33.0 ±0.5 inches water	22 ±1	---	Arm[1/]	Arm[1/]
All go	40.0 ±0.5 inches water	---	22 ±1	Arm[1/]	Arm[1/]
No go	25.0 ±0.5 inches water	24 ±1	28 ±1	Remain safe	Remain safe
Timing	2.0 ±0.1 lb/in^2	24 ±1	28 ±1	11.0 ±1.5 seconds[2/]	11.75 ±2.25 seconds[2/]
Burst	10.0 ±0.3 lb/in^2	24 ±1	28 ±1	Arm[1/]	Arm[1/]

[1/] Rotor fully rotated and locked.

[2/] Rotor fully rotated and locked with time measured as shown on Figure 3.

3.2.1.3 Seal. At the contractor's facility, the leakage rate of the fuze shall not exceed 5 x 10^{-5} standard atmosphere cubic centimeters per second (std atm cm^3/s) when measured within 4 hours after pressurizing and sealing. Anytime after shipment from the contractor's facility, the leakage rate of the fuze shall not exceed 5 x 10^{-4} std atm cm^3/s.

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3.2.1.4 Safety determination. The fuze shall be in the safe condition (see 6.3.1) before and after the capacitor is discharged into a firing cable by application of 28 ±1 volts direct current (Vdc) and 5.6 microfarads (μF) minimum across pins 12(+) and 8(-) of the connector. No explosive component shall initiate.

3.2.2 Environmental conditions. The fuze shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the following environments.

3.2.2.1 Transportation vibration-temperature. The fuze shall meet the requirements of MIL-STD-331, Test 119, Procedure 2, and shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.4.5.1.

3.2.2.2 Shock. The fuze shall meet the requirements of MIL-STD-202, Method 213, Test Condition J, except the amplitude shall be 15 ±1 gravity units (g). The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.4.5.2.

3.2.2.3 Temperature and humidity. The fuze shall meet the requirements of MIL-STD-331, Test 105, either method, and shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.4.5.3.

3.2.2.4 Captive flight vibration (safe condition). While in the safe condition (see 6.3.1), the fuze shall meet the requirements of MIL-STD-810, Method 514, Procedure V, Part 2, Curve AE, except random vibration shall be applied for 2 hours minimum along each of the three mutually perpendicular axes for a total of 6 hours. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.4.5.4.

3.2.2.5 Missile flight vibration (armed condition). While in the armed condition (see 6.3.2), the fuze shall meet the requirements of MIL-STD-810, Method 514, Procedure V, Part 2, spectral density of 0.05 gravity unit squared per Hertz (g^2/Hz), except random vibration shall be applied for 10 minutes minimum along each of the three mutually perpendicular axes for a total of 30 minutes. The fuze shall remain armed, explosive components shall not initiate, and the switches or associated components shall not fail in any way as to interfere with initiation of the leads during and after exposure to the environment of 4.4.5.5.

3.2.3 Environmental safety.

3.2.3.1 Jolt. The fuze shall meet the requirements of MIL-STD-331, Test 101, Procedure II, and shall be safe to handle, store, and transport after exposure to the environment of 4.4.6.1. During exposure,

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no explosive component shall initiate. After exposure, the fuze shall fail to arm or, if arming occurs, the fuze shall arm in not less than the minimum time specified in Table V.

3.2.3.2 Forty-foot drop. The fuze shall meet the requirements of MIL-STD-331, Test 103, Procedure 1, and the requirements specified below after exposure to the environment of 4.4.6.2:

a. No explosive components beyond the detonator shall have initiated.

b. No loose explosive powder shall be present in the housing (Drawing 642AS5551).

c. The lead rotor (Drawing 642AS5422) shall be in the safe position (see 6.3.1) and unable to be turned to the armed position (see 6.3.2).

d. The piston shaft (Drawing 642AS5495) shall be in the safe position (see 6.3.1) and locked by the piston lock (Drawing 642AS5538) and the solenoid cam (Drawing 642AS5334).

e. The fuze shall be safe to handle and dispose of.

3.2.4 Explosive train.

3.2.4.1 Static detonator safety. The fuze shall meet the requirements of MIL-STD-331, Test 115. There shall be no evidence of detonation, burning, charring, scorching, melting, or rupturing of the explosive leads when the detonators are initiated with the arming rotor in the safe position (see 6.3.1). Smudging of the surfaces of the leads, due to deposits resulting from initiation of the detonators, is acceptable. The fuze housing shall not rupture, and any particles of explosive output ejected from the fuze housing through the viewing window shall not be hazardous to personnel or equipment.

3.2.4.2 Explosive-train output. The explosive train of the fuze, in both the instantaneous mode (see 6.3.5) and the delay mode (see 6.3.6), shall produce an average depth of dent of 0.030 inch minimum when armed and detonated on a steel dent block (see Figure 4).

3.2.5 Visual. The fuze shall meet the following requirements:

a. There shall be no evidence of external damage, such as cracks, broken windows, dents, or other blemishes, which may interfere with functioning or fit at the next assembly.

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b. The rotor shall be in the safe position (see 6.3.1), as evidenced by a white letter "S" on a green background centered in the window located on the fuze housing.

c. The safety inspection tag (Drawing 1186AS303) shall be properly signed by both the contractor and Government inspectors, and securely attached to the fuze housing by a lead and wire seal in accordance with MS51938-5, Type I, to indicate that a successful safety test (see 3.2.1.4) has been performed. The contractor's identification symbol shall be pressed into the lead seal.

3.3 First article (preproduction sample). Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample of 30 fuzes to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample shall be at the risk of the contractor.

3.4 Construction. The construction of the fuze shall meet the requirements specified herein and on the drawings and documents listed on DL 1186AS300.

3.4.1 Production drawings. The fuze shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on DL 1186AS300.

3.4.2 Standards of manufacture.

3.4.2.1 Product cleanliness. Product cleanliness shall meet the requirements of MIL-STD-1246, Level 200. No oils or solid lubricants of any kind shall be used on any part of the fuze after final cleaning of components, except as specified on the drawings or herein.

3.4.2.2 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.4.3 Test equipment calibration. When specified in the contract or purchase order, test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. The calibration plan shall be prepared in accordance with the Data Item Description (DID) cited in 6.2.2 and shall be approved by the Government prior to use.

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3.4.4 Test plan and test procedures. When specified in the contract or purchase order, the contractor shall prepare a test plan and test procedures in accordance with the DID cited in 6.2.2. The test plan and test procedures shall be approved by the Government prior to use.

3.4.5 Fuze inspection record. When specified in the contract or purchase order, an inspection record, as shown on Figure 5 and prepared in accordance with the DID cited in 6.2.2, shall be properly filled in for each fuze produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

3.4.6 Workmanship. The fuze, including all parts and accessories, shall be manufactured in a thoroughly workmanlike manner in accordance with the applicable drawings and documents specified herein and on DL 1186AS300. Workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

3.4.6.1 Soldering. Preparation, procedures, and requirements for soldering of electrical connections shall be in accordance with WS-6536. Mechanical soldering shall be in accordance with MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Preproduction sample inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

1. Individual unit acceptance inspections (see 4.2.3).

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4.1.2 Special tests and examinations.

4.1.2.1 Preproduction sample. The preproduction sample of 30 fuzes shall be subjected to the individual unit acceptance inspections of Figure 6 (see 4.2.3). The preproduction sample shall then be subjected to the inspections of Figures 7 and 8 in the sequence shown. The inspections of Figure 8 shall be performed at an independent facility designated in the contract or purchase order (see 6.2.1).

4.1.2.1.1 Preproduction sample acceptance and rejection criteria. When subjected to the inspection sequences of Figures 7 and 8, each fuze of the preproduction sample shall meet the applicable performance requirements of Section 3 on the first and only trial. Defects are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified on the drawings. The preproduction sample shall be rejected if:

a. One or more CRITICAL defects are found, or

b. Two or more MAJOR defects are found.

NOTE: Although one MAJOR defect does not cause rejection of the preproduction sample, appropriate corrective action shall be taken to prevent recurrence prior to production.

4.2 Quality conformance inspections. The tests and examinations of 4.4 shall be performed to determine that all the requirements of Section 3 have been met. Tests shall be performed in accordance with the test plan and test procedures which has been approved by the Government (see 3.4.4).

NOTE: The contractor may, prior to forming and submitting a lot for acceptance, subject the completely assembled fuze to the inspections of 4.4.1 through 4.4.4, 4.4.8, 4.4.9, and 4.4.10 to the extent deemed necessary.

4.2.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain not less than 51 nor more than 280 fuzes, including those required for destructive tests (see 6.2.1). An inspection lot, as defined in MIL-STD-105, shall consist of completed fuzes fabricated under the same conditions and offered for inspection at the same time.

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4.2.1.1 Serial numbers. Unless otherwise specified in the contract or purchase order, all fuzes shall be assigned a Government serial number in order of consecutive assembly (see 6.2.1). After assignment of a Government serial number to each fuze, the inspection lot shall be formed and identified by lot number.

4.2.2 Quality conformance inspection sequence. Prior to formation of the lot, the fuze shall be subjected to the individual unit acceptance inspections of Figure 6 (see 4.2.3). After assignment of serial numbers, and after the inspection lot has been formed and identified by number, the lot shall be subjected to the inspections of Figure 9. After completion of the inspections of Figure 9, 15 fuzes shall be randomly sampled from the inspection lot in accordance with MIL-STD-105 and subjected to the inspections of Figure 8 at an independent facility designated in the contract or purchase order (see 6.2.1). The remainder of the lot shall be held in bond pending acceptance of the 15-fuze sample.

4.2.2.1 Quality conformance acceptance and rejection criteria. Defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified on the drawings. Acceptance and rejection shall be based on the following:

a. For the inspections of Figure 6, all parts, subassemblies and assemblies found to be defective shall be rejected on an individual basis and shall not be used in further assembly of the fuze (see 4.2.3).

b. For the inspections of Figure 9:

1. Where 100 percent inspection is required, all defective fuzes shall be rejected.

2. Where sampling is required for lot sizes of 51 to 159 fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Single Sampling Plan, Tightened Inspection, sample size code letter H, AQL of 0.40 percent defective. For lot sizes of 160 to 280 fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Double Sampling Plan, Normal Inspection, sample size code letter K, AQL of 0.40 percent defective.

c. For the inspections of Figure 8, the failure of any sample fuze to meet the applicable requirements of Section 3 shall cause rejection of the lot.

4.2.2.1.1 Counting of defects. When counting defects found in fuzes to determine acceptability of the lot, fuzes with multiple defects shall be counted as a single defect, using the highest classification of the defects found. For example, when a fuze has both a CRITICAL and MAJOR defect, it shall be counted as a CRITICAL defect.

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4.2.3 Individual unit acceptance inspections. Individual unit acceptance inspections shall be performed during the manufacture of the fuze in accordance with Figure 6. Parts, subassemblies, or assemblies found to be defective during the inspection sequence shall not be used in further assembly of the fuze.

4.3 Test equipment and conditions.

4.3.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. All test equipment shall be calibrated as specified in 3.4.3.

4.3.2 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government (see 3.1.2), are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.3.1 and are subject to approval by the procuring activity.

4.3.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. Unless otherwise specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.3.4 Test environment. Unless otherwise specified herein, the inspection methods of 4.4 shall be performed under the conditions specified in 3.4.2.2.

4.3.5 Resetting. Before detachment from the test set, all fuzes shall be reset to the safe position (see 6.3.1) after any test where arming occurs, except for those fuzes which must be left in the armed position (see 6.3.2) for subsequent tests. Although failure of a fuze to reset to the safe position does not affect acceptance of a lot, any fuze which cannot be reset shall be rejected. Unless otherwise specified in the contract or purchase order, disposition of armed fuzes that repeatedly fail to reset to the safe position shall be at the discretion of the cognizant demolition officer (see 6.2.1). The unsafe condition shall be indicated by securely attaching an unsafe condition tag as shown on Figure 10.

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4.4 Methods of inspection.

WARNING

Unless otherwise specified, verify that the fuze is in the safe condition (see 6.3.1) before removing the fuze from the test set at any time during any test sequence. Discharge the firing capacitor by shorting pins 5 and 8 together for 15 seconds minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

4.4.1 Electrical. The tests of 4.4.1.1 and 4.4.1.2 shall be conducted on fuzes during and after assembly as specified on the applicable figures. Resistance measurements shall be performed using the test set (see 3.1.2 and 3.2.1.1), or standard test equipment that meets the requirements of 4.3.1. Required test data shall be recorded on the inspection record (see Figure 5).

4.4.1.1 Resistance (safe condition). To determine conformance to 3.2.1.1.1, verify the resistance requirements of Table II with the fuze in the safe condition (see 6.3.1). The high pot (HP) tests shall be performed at 500 Vdc ±10 percent. The resistance shall be within the limits specified in the table.

4.4.1.2 Resistance (armed condition). To determine conformance to 3.2.1.1.2, verify the resistance requirements of Tables III and IV with the fuze in the armed condition (see 6.3.2). The tests for configuration C (see Table I) of Table IV shall be performed during the environmental inspection sequence of Figure 8. The resistance shall be within the limits specified in the tables.

4.4.2 Mechanical operability. To determine conformance to 3.2.1.2, verify the mechanical operability requirements of Table V using the test set (see 3.1.2 and 3.2.1.1), or standard test equipment with a two-channel (minimum) recorder and acoustic pickup, which meets the requirements of 4.3.1. Timing shall be determined from the output record chart as shown on Figure 3. Record test results on the inspection record (see Figure 5).

NOTE: An acoustic pickup shall be used as an audio aid to verify fuze arming and resetting (see 4.3.5).

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4.4.3 Seal. To determine conformance to 3.2.1.3, test the fuze in accordance with MIL-STD-202, Method 112, Test Condition C, Procedure IV. Trichlorotrifluoroethane (MIL-T-14757) or isopropyl alcohol (TT-I-735, grade optional) may be used as alternative immersion fluids for gross leak testing. At the contractor's facility, the fuze shall be tested within 4 hours after pressurizing and sealing and the results shall be recorded on the inspection record (see Figure 5).

4.4.4 Safety determination. To determine conformance to 3.2.1.4, test the fuze as follows:

a. If required, reset the fuze to the safe condition (see 6.3.1) using the test set (see 4.3.5). Discharge the firing capacitor by shorting pins 5 and 8 together for 15 seconds minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

b. Visually verify that the fuze is in the safe condition (see 6.3.1), as evidenced by a white letter "S" on a green background centered in the window located on the fuze housing.

WARNING

Do not proceed with the test unless the fuze is in the safe condition (see 6.3.1). The following test will cause initiation of the explosive train if the fuze is not in the safe condition.

c. Attach a firing cable to pins 12(+) and 8(-) of the connector.

d. Place the fuze in a safety chamber.

e. Connect a firing cable to a capacitor discharge system capable of delivering a minimum of 20,000 ergs to the fuze (28 ±1 Vdc and 5.6 μF).

f. Discharge the capacitor into the firing cable.

g. Run a continuity check between pins 12(+) and 8(-) to determine if the fuze is safe to handle (open indication on the 1-kilohm (kohm) scale of the test set).

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h. Remove the fuze from the safety chamber. Visually verify that the fuze is in the safe condition (see 6.3.1) as specified in b above.

i. Fill in, sign, and securely attach a safety inspection tag (Drawing 1186AS303) to the housing using a lead and wire seal (see 3.2.5c).

j. Record the test results on the inspection record (see Figure 5).

4.4.5 Environmental. All fuzes of the preproduction sample, or a randomly selected sample of 15 fuzes from an inspection lot, shall be subjected to the following environmental tests in accordance with Figure 8 at an independent test facility designated in the contract or purchase order (see 6.2.1). Any fuze subjected to these tests and the tests of 4.4.6 and 4.4.7 shall be considered destroyed and shall not be returned to the lot nor forwarded for service use, but shall be labeled appropriately and held for disposal as specified in the contract or purchase order (see 6.2.1). During these tests and the tests of 4.4.6 and 4.4.7, initiation of any explosive component, except those which are initiated intentionally as part of a test, shall be considered a CRITICAL defect and shall cause rejection of the lot from which the fuze was drawn. Throughout the tests, the fuze shall be in the safe condition (see 6.3.1) unless otherwise specified herein or on the drawing.

CAUTION

The explosive components of the fuze are highly sensitive and hazardous; proper safety precautions must be observed during handling (see 6.4).

4.4.5.1 Transportation vibration-temperature. To determine conformance to 3.2.2.1, mount the fuze to the vibration fixture (Drawing SA 498019). Test the fuze in accordance with MIL-STD-331, Test 119, Procedure 2.

NOTE: For the vibration and shock tests specified herein, the test fixtures shall be mounted to a vibration table or shock machine and the mounting bolts shall be tightened to a torque of 25 ±1 foot-pounds.

4.4.5.2 Shock. To determine conformance to 3.2.2.2, mount the fuze to the shock fixture (Drawing SA 2876062; alternate, vibration fixture of Drawing SA 498016). Test the fuze in accordance with MIL-STD-202, Method 213, Test Condition J, except that the amplitude shall be 15 ±1 g.

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4.4.5.3 Temperature and humidity. To determine conformance to 3.2.2.3, test the fuze in accordance with MIL-STD-331, Test 105, either method. The connector shall be thoroughly cleaned and dried after the test.

4.4.5.4 Captive flight vibration (safe condition). To determine conformance to 3.2.2.4, mount the fuze to the vibration fixture (Drawing SA 498016). Test the fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, Curve AE, except random vibration shall be applied for 2 hours minimum along each of the three mutually perpendicular axes for a total of 6 hours.

4.4.5.5 Missile flight vibration (armed condition). To determine conformance to 3.2.2.5, mount the armed fuze to the vibration fixture (Drawing SA 498016). Test the fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, spectral density of 0.05 g^2/Hz. Random vibration shall be applied for 10 minutes minimum along each of the three mutually perpendicular axes for a total of 30 minutes. Throughout the test, the fuze shall be electrically monitored to determine whether it remains in the armed condition (see 6.3.2).

4.4.6 Environmental safety. The conditions of 4.4.5 shall apply for the tests of 4.4.6.1 and 4.4.6.2.

4.4.6.1 Jolt. To determine conformance to 3.2.3.1, mount the fuze to the jolt fixture (Drawing SA 498017). Test the fuze in accordance with MIL-STD-331, Test 101, Procedure II. After jolt, test the fuze in accordance with 4.4.2.

4.4.6.2 Forty-foot drop. To determine conformance to 3.2.3.2, mount the fuze to the forty-foot drop fixture (Drawing SA 498018). Test the fuze in accordance with MIL-STD-331, Test 103, Procedure 1. When only three fuzes are to be dropped (see Figure 8), the drop positions shall be vertical (connector up), horizontal, and 45 degrees from vertical (connector down).

4.4.7 Explosive train. The conditions of 4.4.5 shall apply for the tests of 4.4.7.1 and 4.4.7.2.

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4.4.7.1 Static detonator safety. To determine conformance to 3.2.4.1, test the fuze in accordance with MIL-STD-331, Test 115, and as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Verify that the fuze meets the safe condition requirements of 4.4.4a and b.

b. For explosive output witness, wrap the fuze with one layer of polyester pressure-sensitive tape (L-T-100, Type I), centered over the viewing window in the housing.

c. Attach a shorting plug (all pins shorted) to the connector.

d. Machine off the welded seal around the flange of the fuze and remove the internal assembly from the housing. Place the assembly behind a protective shield.

e. Modify terminal board TB1 in the following sequence and as shown on Figure 11.

1. Remove diode CR7 from terminal board TB1 between terminals 3 and 4.

2. Remove diode CR4 from terminal board TB1 between terminals 2 and 5.

3. Remove thermofit insulation from terminal board TB1, terminals 3 and 4. Remove solenoid wires and clean terminals.

4. Add 22-gauge jumper wire from terminal 1 to terminal 6.

5. Add 22-gauge jumper wire from terminal 3 to terminal 5.

6. Add 22-gauge jumper wire from terminal 2 to terminal 4.

f. Reinsert the internal assembly into the housing. Clamp the internal assembly and the housing together in such a manner to prevent separation and movement during the test.

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g. Place the fuze into a firing chamber capable of containing 2.2 grams of CH-6 (MIL-C-21723) explosive output.

h. Remove the shorting connector and connect the static detonator safety test cable to pins 10(+) and 8(-).

i. Initiate the detonators using a Dupont Blasting Machine Condensor Discharge CD-12-1 (or equivalent equipment with a 100-μF capacitor charged to 130 Vdc).

j. Remove the fuze from the firing chamber, and detach the fuze from the test setup. Inspect the housing for evidence of rupture, and the witness tape of b above for evidence of perforation. Then, disassemble the fuze and inspect the interior for conformance to 3.2.4.1.

4.4.7.2 Explosive-train output.

WARNING

The explosive-train output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 6.4).

The explosive-train output test consists of initiating the fuze on a steel dent block (see Figure 4) and measuring the depth of the dents to determine the output.

4.4.7.2.1 Instantaneous mode. To determine conformance to 3.2.4.2, modify and test the armed fuze as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Arm the fuze using the test set.

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b. Before handling the fuze, discharge the firing capacitor by shorting pins 5 and 8 together for 15 seconds minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

c. Verify that the fuze is in the armed condition (see 6.3.2).

d. Place the armed fuze in a firing chamber capable of containing 2.2 grams of CH-6 (MIL-C-21723) explosive output, with the output end aligned on top of the dent block (see Figure 4).

e. Attach a firing cable to pins 8(+) and 12(-). Close the firing chamber.

f. Initiate the detonators using a Dupont Blasting Machine Condenser Discharge CD-12-1 (or equivalent equipment with a 100-μF capacitor charged to 130 Vdc).

g. Measure the continuity between pins 8(+) and 12(-) to determine if the fuze is safe to handle (open indication on the 1-kohm scale of the test set).

h. Remove all foreign material from the dent block. (There should be two dents in the dent block.)

i. Place the gage probe of a dial-indicator height gage in the deepest part of the first dent and zero the gage indicator.

j. Remove the probe from the dent and take gage readings at two points, each approximately 0.125-inch from the opposite edge of the dent block as shown on Figure 4. Record the average of the two readings.

k. Repeat i and j for the second dent. The average depth of each dent shall conform to 3.2.4.2.

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4.4.7.2.2 Delay mode. To determine conformance to 3.2.4.2, modify and test the armed fuze as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Repeat 4.4.7.2.1a, b, and c.

b. Place the explosive-train drop fixture (Drawing SA 497753) into a firing chamber capable of containing 2.2 grams of CH-6 (MIL-C-21723) explosive output.

c. Place the dent block (Figure 4) in a booster cup in accordance with Drawing 2049577 (or equivalent). Screw the booster cup to the output end of the armed fuze.

d. Install the armed fuze in the drop fixture.

e. Attach a firing cable to pins 8(+) and 10(-). Close the firing chamber. Apply 28 ±1 Vdc to charge the firing capacitor.

f. After charging 30 seconds minimum, pull the fuze release drop wire on the test fixture. (Upon impact, the fuze firing circuit should close, initiating the delay detonators.)

g. Repeat 4.4.7.2.1g through k.

4.4.8 Visual inspection. Visually inspect the fuze to determine conformance to 3.2.5 and the general requirements of Section 3.

4.4.9 Drawing inspections. To determine conformance to 3.4.1, perform drawing inspections on parts, subassemblies, and assemblies of the fuze during manufacture, and on the completely assembled, sealed, and pressurized fuze. The sampling plans and acceptance criteria shall be as specified on the drawings.

4.4.10 Packaging inspection. Prior to shipment, inspect the fuze to determine conformance to the requirements of Section 5.

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5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on DL 1186AS311.

5.1.1 Level C. Fuzes shall be packaged in accordance with the drawings listed on DL 1186AS311 and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on DL 1186AS311.

5.2.1 Level C. Fuzes, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and shipping containers shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with the drawings listed on DL 1186AS311 and MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The fuze is intended for use as a major component of the Tomahawk Cruise Missile.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Number, title, and date of this specification.

b. Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

c. Government-loaned property. The contracting officer should arrange to loan the property cited in 3.1.2.

d. First article. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 30 fuzes as specified in 3.3 and 4.1.2.1. The

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contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

e. Responsibility for inspection, if other than as specified in 4.1.

f. Inspection facility, if other than as specified in 4.1.

g. Independent facility to perform the inspections of Figure 8 (see 4.1.2.1, 4.2.2, and 4.4.5).

h. Inspection lot size, if other than as specified in 4.2.1.

i. Assignment of serial numbers, if other than as specified in 4.2.1.1.

j. Government final inspection gage drawings (see 4.3.3).

k. Disposition of armed fuzes which repeatedly fail to reset to safe position, if other than as specified in 4.3.5.

l. Disposal of fuzes subjected to inspections of Figure 8 (see 4.4.5).

m. Special marking (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.4.3	Calibration Plan	UDI-T-23742A
3.4.4	Test Plan and Test Procedures	DI-T-5204
3.4.5	Inspection Record	UDI-M-20404

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 Definitions.

6.3.1 Safe condition and safe position. Visually, the fuze is considered to be in the safe condition or safe position when a white letter "S" on a green background is centered in the viewing window located in the housing. Electrically, the fuze is considered to be in the safe condition or safe position when no explosive components of the fuze initiate during the test of 4.4.4.

6.3.2 Armed condition and armed position. Visually, the fuze is considered to be in the armed condition or armed position when a white letter "A" on a red background is centered in the viewing window located in the housing. Electrically, the fuze is considered to be in the armed condition or armed position when, as measured on the test set, the limits of Tables III and IV are met under the conditions specified in the tables.

6.3.3 Safe. Safe is defined as the ability of the fuze to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the fuze to maintain its safety features in a condition which will permit disposal without injury to personnel.

6.3.4 Operable. Operable is defined as the ability of the fuze, with specified inputs, to perform to completion its specified function and sequence, and produce the specified outputs within the required operational period or at the specified point of time.

6.3.5 Instantaneous mode. The instantaneous mode is defined as the instantaneous initiation of the MARK 71 MOD 1 Detonators when a target is sensed. The instantaneous mode was designed to accommodate a proximity fuze which can be incorporated in the missile as a backup fuzing system.

6.3.6 Delay mode. The delay mode is defined as the initiation of the MARK 85 MOD 0 Detonators after a specified time delay following impact.

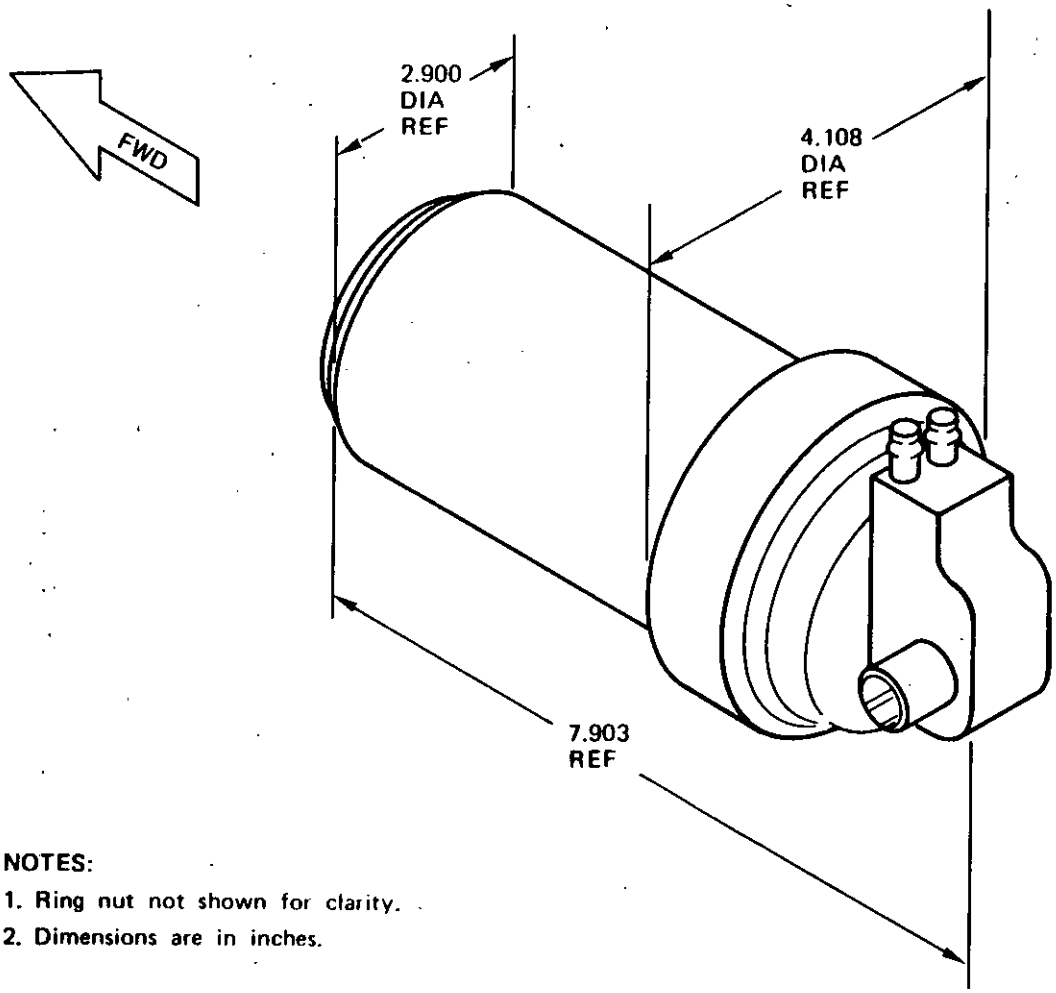
6.4 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

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NOTES:

1. Ring nut not shown for clarity.
2. Dimensions are in inches.

FIGURE 1. FMU-138/B Guided Missile Fuze general configuration.

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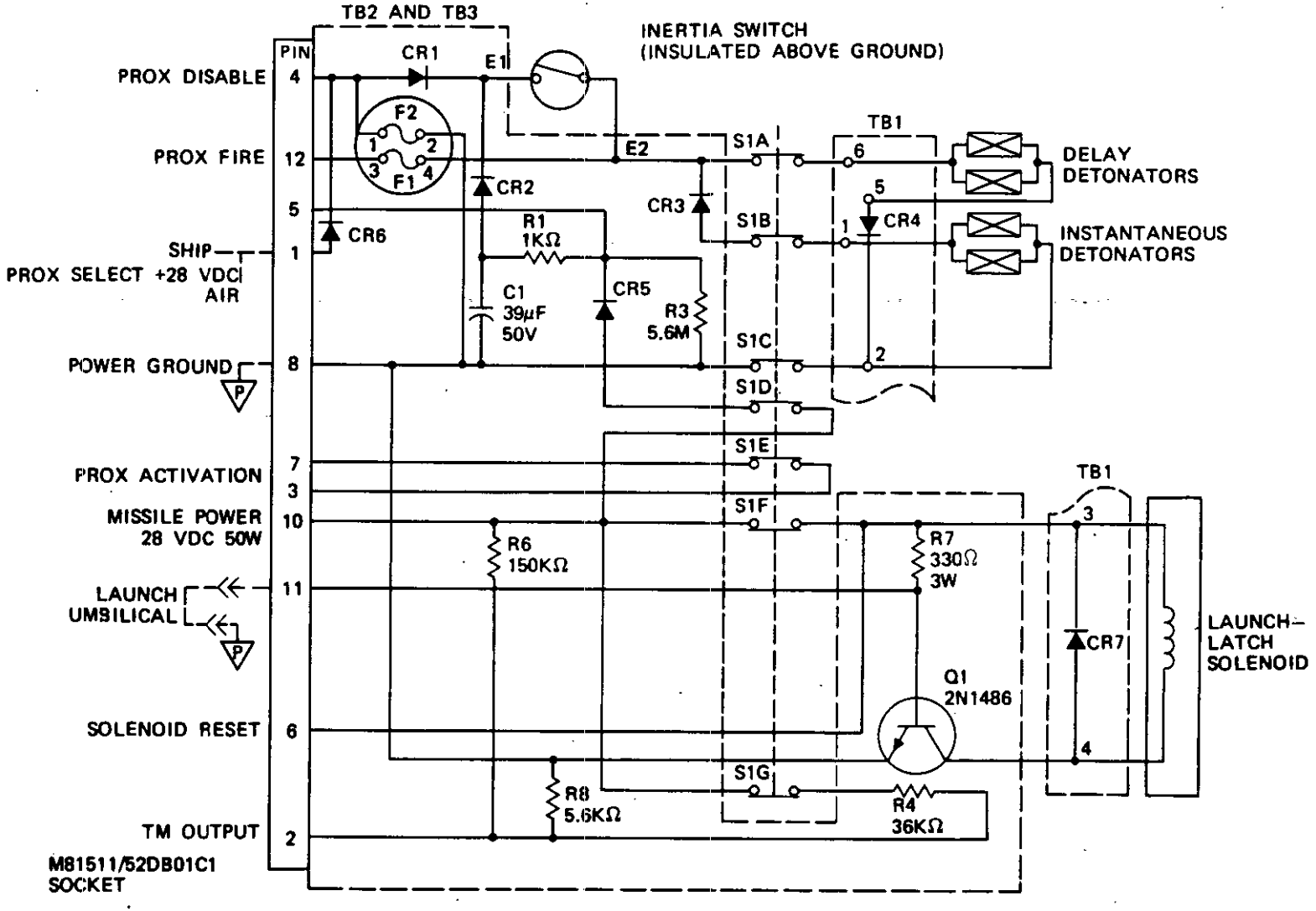


FIGURE 2. Electrical schematic.

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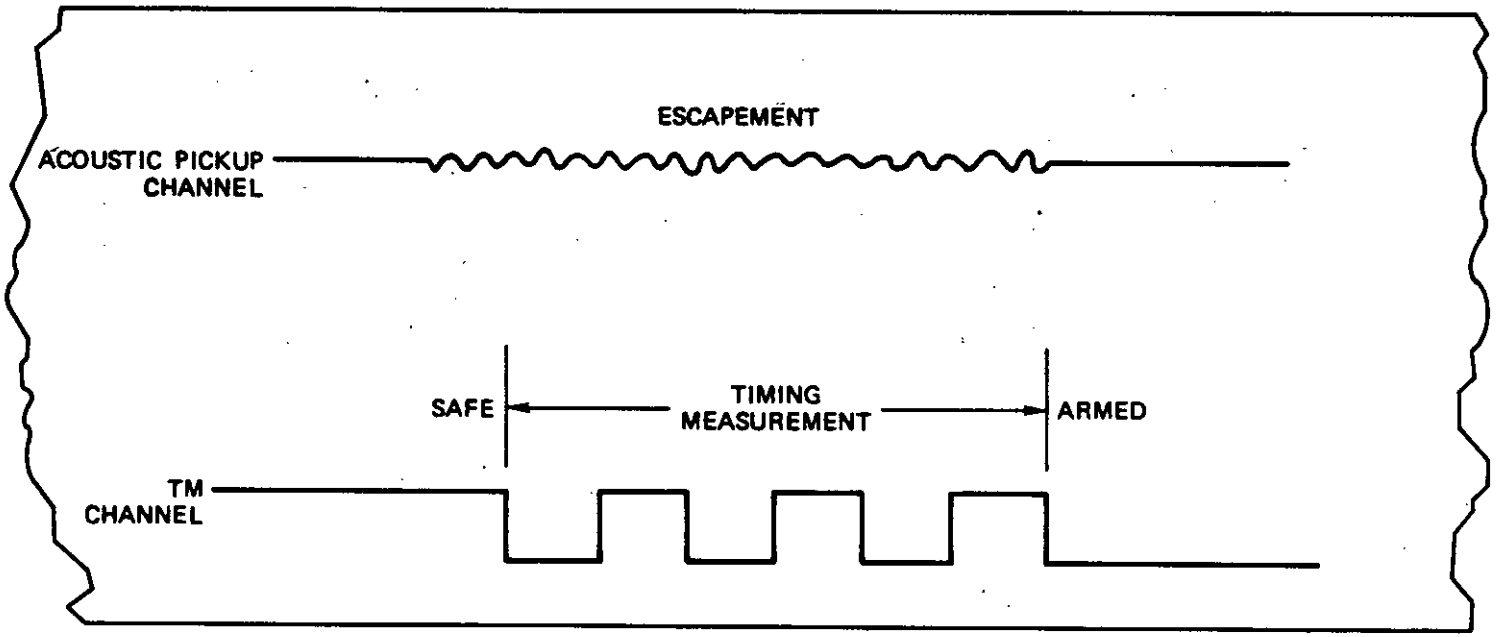
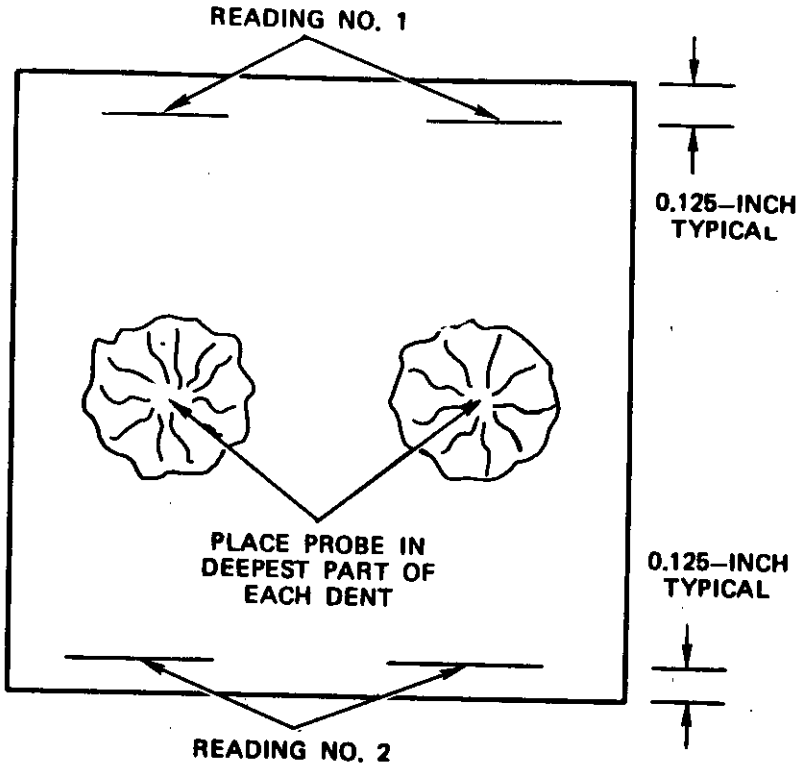


FIGURE 3. Sample of test set fuze timing output record chart.

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NOTES:

1. Dent block shall be steel, cold-finished bar in accordance with ASTM A 108, Grade 1018 or 1020, Rockwell hardness B70 to B95, surface condition as rolled. Size, 1.750 ±0.125 inches square, 1.250 ±0.125 inches thick. Block shall be clean and free from rust and burrs.
2. Dent block shall be placed inside a booster cup in accordance with Drawing 2049577 (or equivalent) for the test of 4.4.7.2.2.

FIGURE 4. Dent block and dent measurement.

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INSPECTION RECORD
FMU-139/B GUIDED MISSILE FUZE

CONTRACT NO. _____ MANUFACTURER _____

PREPRODUCTION SAMPLE ☐
INSPECTION LOT ☐
REWORK ☐

S/N _____ LOT NO. _____

INSPECTOR _____ DATE _____

ELECTRICAL RESISTANCE

SAFE CONDITION

TEST	CONFIG A	CONFIG B	CONFIG C	TEST	CONFIG A	CONFIG B	CONFIG C
HP1				S6			
HP2				S7			
S1				S8			
S2				S9			
S3				S10			
S4				S11			
S5				S12			

ARMED CONDITION

TEST	CONFIG A	CONFIG B
A1		
A2		
A3		
A4	NA	
A5		

CHECKED BY _____
DATE _____

DRAWING INSPECTIONS

DWG 842AS5422 VERIFY NOTE 5
CHECKED BY _____ DATE _____

DWG 1186AS300 VERIFY NOTE 12
CHECKED BY _____ DATE _____

DWG 1188AS217 VERIFY NOTE 12

TEST	OHMS	TEST	OHMS
13		23	
14		26	
22			

CHECKED BY _____ DATE _____

PARTS/SUBASSEMBLIES
VERIFIED BY _____ DATE _____

DWG 1186AS300 — FINAL ASSEMBLY
VERIFIED BY _____ DATE _____

MECHANICAL OPERABILITY

TEST	RESULT	TEST	RESULT
ALL GO		TIMING	
NO GO		BURST	

CHECKED BY _____ DATE _____

SEAL

LEAK RATE _____ STD ATM CM^3/s
CHECKED BY _____ DATE _____

SAFETY DETERMINATION
VERIFIED BY _____ DATE _____

FINAL VISUAL INSPECTION

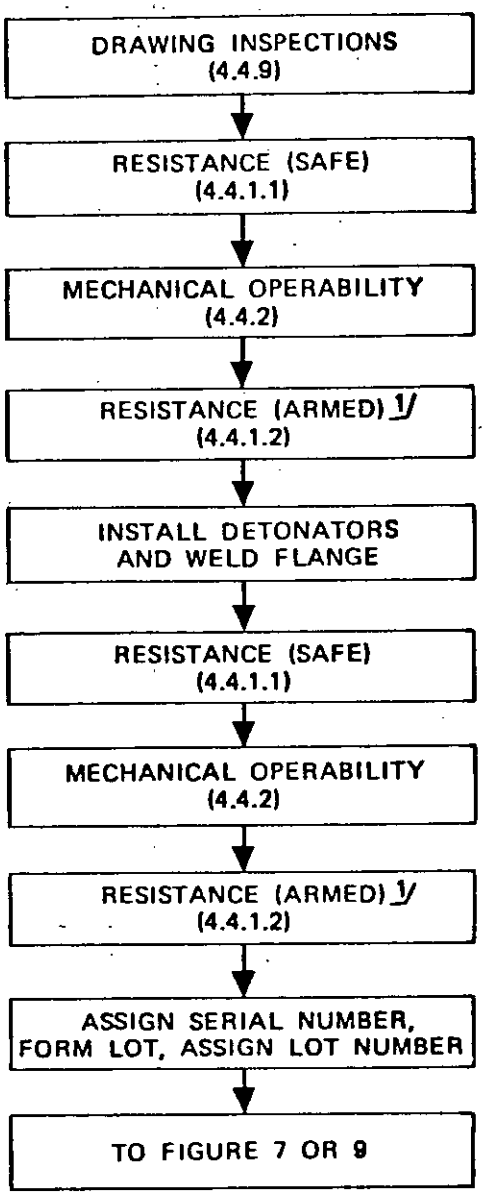
	YES	NO
(1) EXTERNAL DAMAGE	☐	☐
(2) FUZE IN SAFE CONDITION (WHITE "S" VISIBLE THROUGH WINDOW)	☐	☐
(3) SAFETY INSPECTION TAG ATTACHED BY LEAD AND WIRE SEAL, PROPERLY FILLED IN, AND ALL SIGNATURES PRESENT	☐	☐

GOVERNMENT REPRESENTATIVE:

_____ _____
SIGNATURE DATE

FIGURE 5. Sample inspection record.

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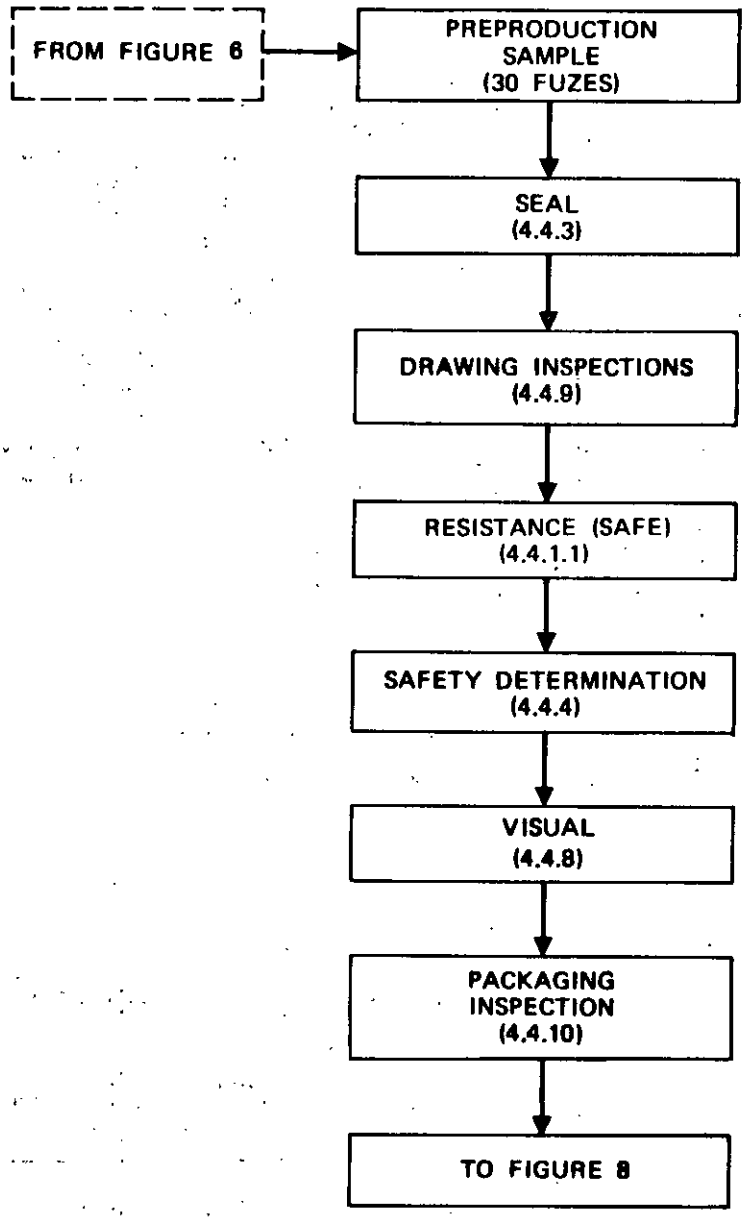
NOTES:

1. Test results shall be recorded on the inspection record of Figure 5.

1/ Reset the fuze to the safe condition after test (see 4.3.5).

FIGURE 6. Individual unit acceptance inspection sequence.

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NOTE: All test data shall be recorded on the inspection record (see Figure 5).

FIGURE 7. Preproduction sample inspection sequence.

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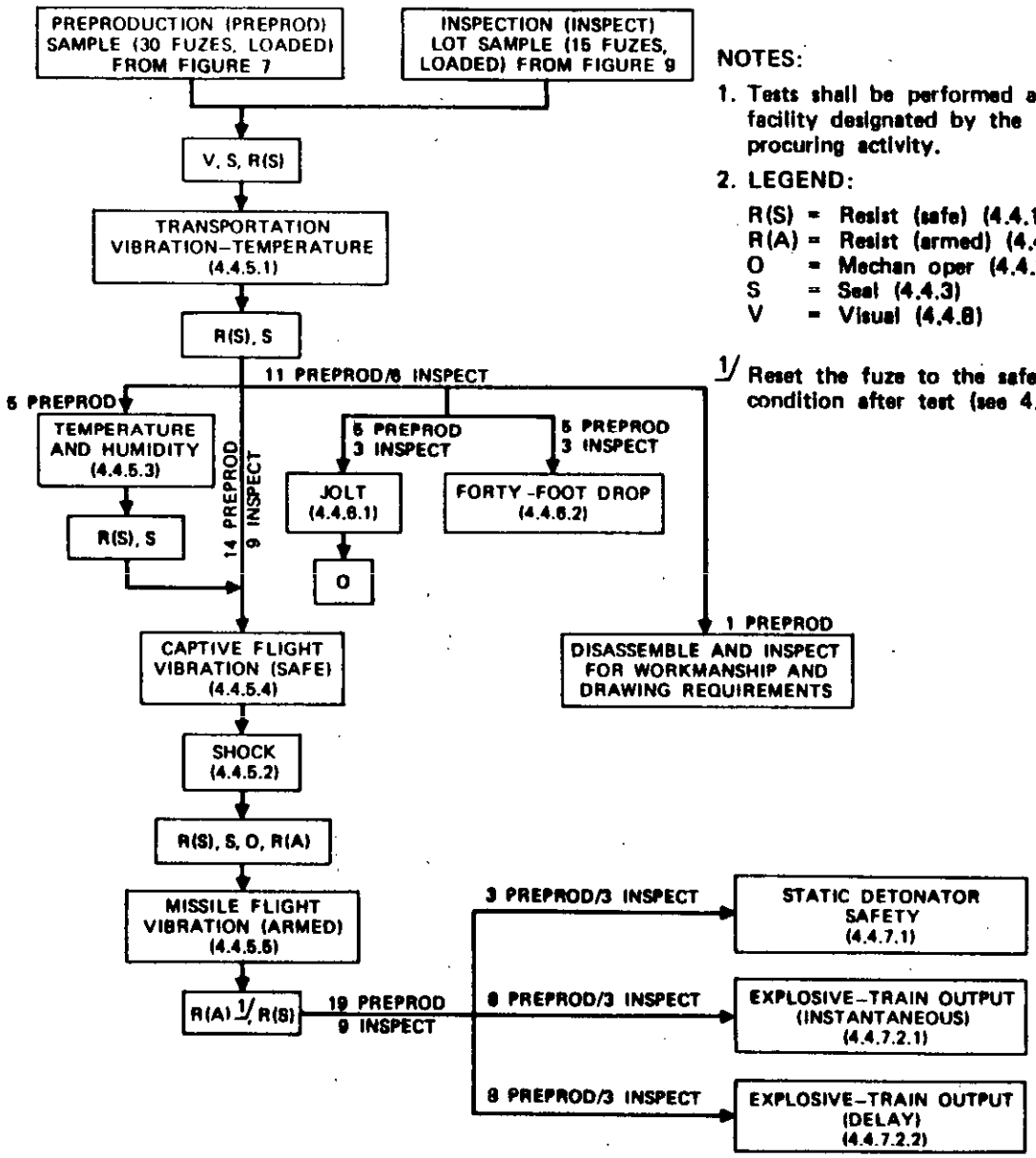
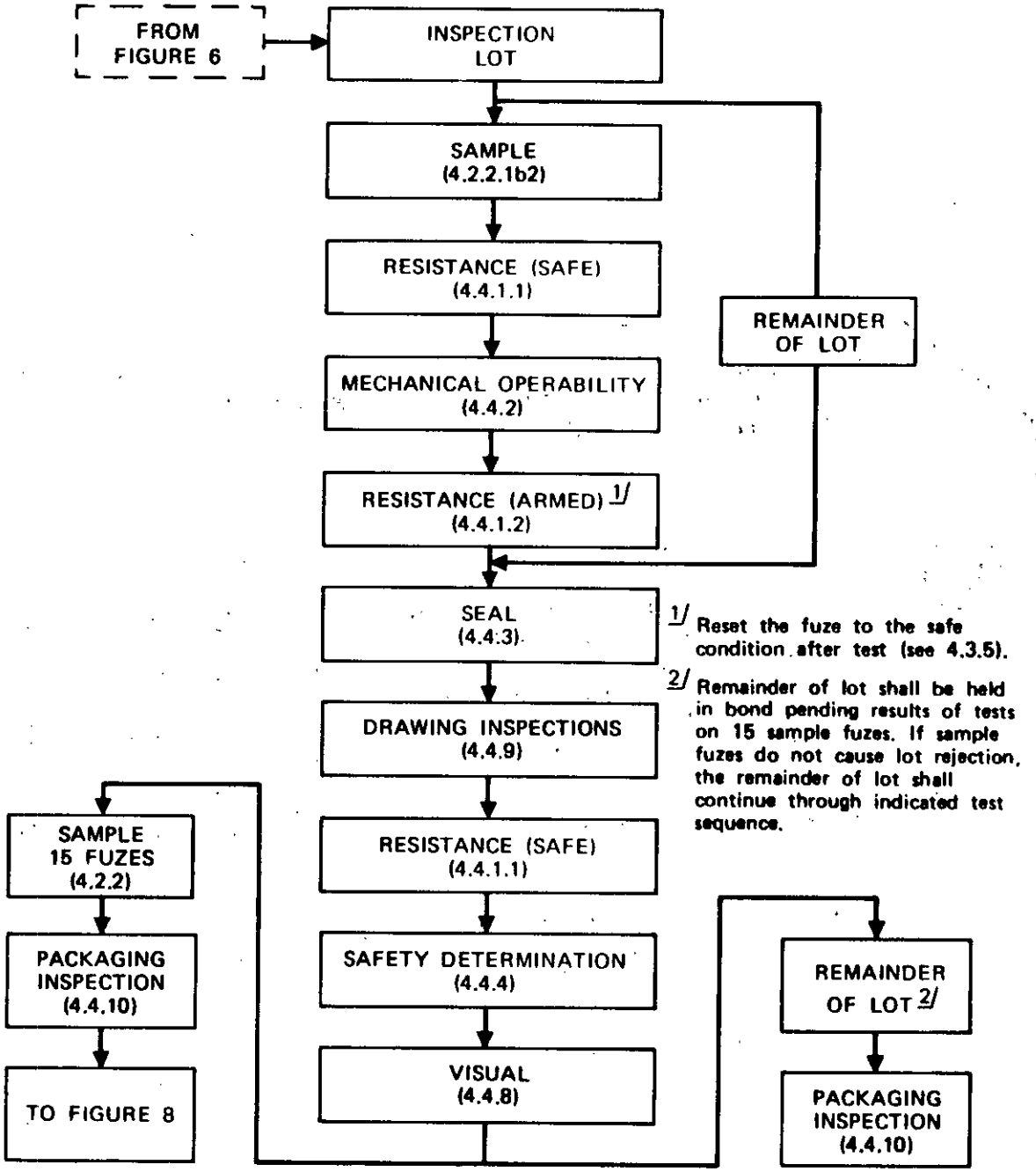


FIGURE 8. Environmental and explosive-train inspection sequence.

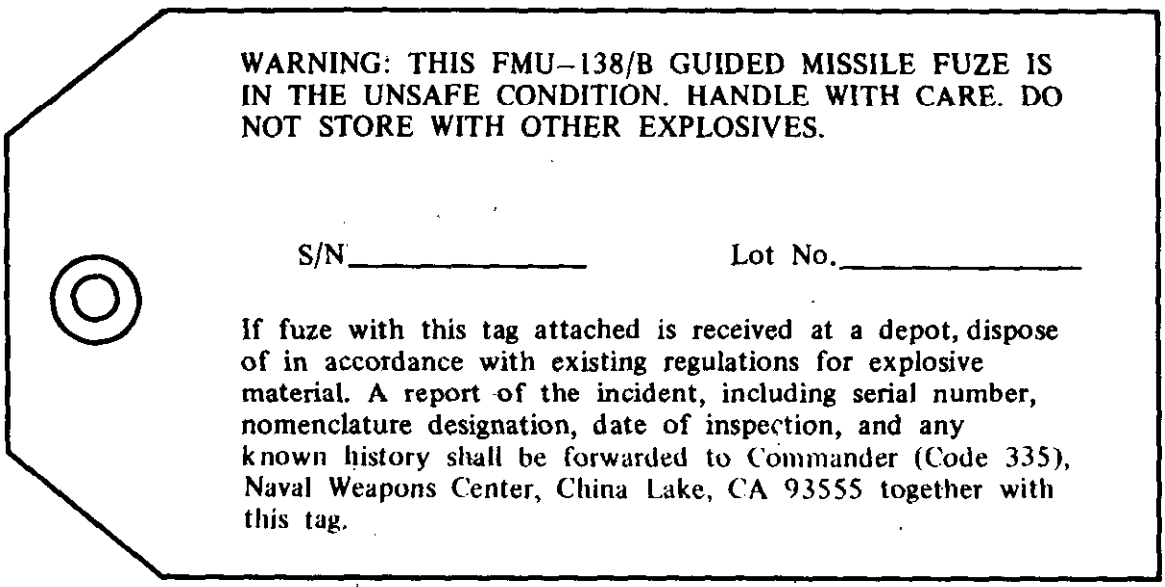
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NOTE: Test data shall be recorded on inspection record (see Figure 6).

FIGURE 9. Inspection lot inspection sequence.

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NOTE: Tag shall be in accordance with UU-T-81, Type B, Grade 15SU, Size 7 or 8, Color red. The required data may be printed or stamped thereon.

FIGURE 10. Unsafe condition tag.

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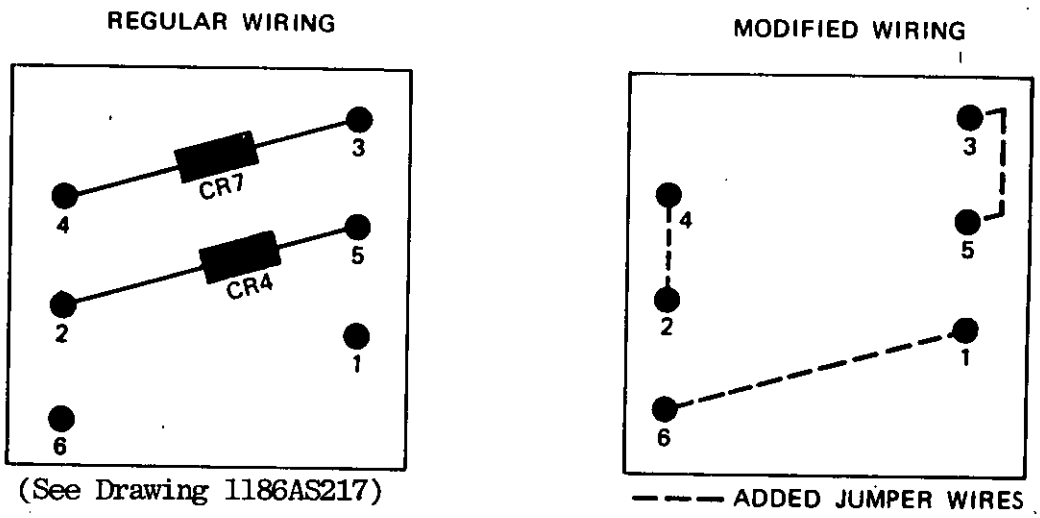


FIGURE 11. Terminal board TB1 modification for static detonator safety test.

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